1. Source tables

EMP

NAME	OFFICE	DEPT	SALARY
Sumith	400	CS	25000
Ravi	220	Econ	30000
Kiran	160	Elec	35000
Sathish	420	Econ	40000

DEPART

DEPT	MAIN OFFICE	PHONE
CS	404	2436789
Econ	200	2378658
Fin	501	2356297
Hist	100	2745897

We want to **link each employee to details about their department** (main office, phone), using the common column DEPT.

2. Join condition

Relational algebra expression:

$$EMP \bowtie_{emp.dept=depart.dept} DEPART$$

Meaning:

- Take every EMP row and every DEPART row.
- Keep a pair only if **EMP.DEPT = DEPART.DEPT**.
- This is a typical **equi- join** because the join predicate is equality on DEPT.
- It is also an **inner join**, so **unmatched** rows from either table are **discarded**.

3. Matching rows step- by- step

Check each EMP row against DEPART:

1. **EMP: Sumith, DEPT = CS**
    - Match with DEPART row where DEPT = CS → (CS, 404, 2436789).
    - Joined row becomes:
      (Sumith, 400, CS, 25000, CS, 404, 2436789).
2. **EMP: Ravi, DEPT = Econ**
    - Match with DEPART row Econ → (Econ, 200, 2378658).
    - Joined row:
      (Ravi, 220, Econ, 30000, Econ, 200, 2378658).
3. **EMP: Kiran, DEPT = Elec**
    - There is **no DEPART row with DEPT = Elec**.
    - Inner join drops this employee completely (no row in result).
4. **EMP: Sathish, DEPT = Econ**
    - Again matches DEPART Econ → (Econ, 200, 2378658).
    - Joined row:
      (Sathish, 420, Econ, 40000, Econ, 200, 2378658).

Notice: DEPART rows Fin and Hist also have no matching employees, so they do **not** appear in the result either.

4. Final join result

The slide shows the result table:

NAME	OFFICE	emp.DEPT	SALARY	Depart.DEPT	MAIN OFFICE	PHONE
Sumith	400	CS	25000	CS	404	2436789
Ravi	220	Econ	30000	Econ	200	2378658
Sathish	420	Econ	40000	Econ	200	2378658

Key points:

- **Kiran** is missing because his department (Elec) has no matching DEPART row.
- **Fin** and **Hist** departments are missing because no employee belongs to them.
- Column **DEPT** appears twice: once as emp.DEPT and once as Depart.DEPT, because equi- join keeps all columns from both tables and does not remove duplicate join attributes. (A natural join would remove one copy.)

5. Intuitive interpretation

You can read the join as:

> "For each employee, look up their department in DEPART; if that department exists, attach its main office and phone; if not, drop that employee from the result."

Thus, the example demonstrates that an **inner equi- join** returns only those employees whose department is defined in the department table, enriched with department details.